ShiftGuard: Risk-Controlled Spatiotemporal Traffic Forecasting under Compound Regime Shifts and Sensor Outages

Anonymous Author[a]

[a] *Anonymous Institution*

Abstract

Accurate short-term traffic forecasting is essential for intelligent transportation systems, yet conventional benchmarks largely assume stable traffic regimes and fully observed sensor networks. Real deployments instead experience compound distribution shifts in which traffic conditions change while spatially and temporally correlated sensors become unavailable. This study presents ShiftGuard, a risk-controlled spatiotemporal forecasting framework that jointly produces point forecasts, calibrated node–horizon prediction intervals, and origin-level abstention decisions. Its principal components are a reliability-conditioned graph–temporal encoder, structured-outage augmentation, a heterogeneity-aware probabilistic head, and prequential group-adaptive conformal calibration. A leakage-controlled evaluation protocol is further introduced to test independent, burst, contiguous-region, and high-centrality hub outages across multiple severities and traffic regimes on METR-LA and PEMS-BAY. The empirical study includes clean testing and 640 confirmatory stress runs over five random seeds. Across the four non-clean outage families and five severity levels, ShiftGuard lowers mean absolute error relative to the strongest locally reproduced transformer from 3.74 to 3.40 mph on METR-LA and from 2.49 to 1.94 mph on PEMS-BAY. Crossed seed–scenario bootstrap intervals for these improvements are $[-0.465, -0.234]$ and $[-0.682, -0.411]$ mph, respectively. At 30% outage severity, the risk controller retains 86% and 98% of forecast origins while restricting empirical origin-level failure to 8% and 2%. Although clean PEMS-BAY accuracy remains 0.027 mph behind STAEformer, the results demonstrate that ShiftGuard provides stronger robustness, calibrated uncertainty, and actionable failure control under realistic compound shifts.

Keywords: traffic forecasting, intelligent transportation systems, spatiotemporal graph neural network, sensor outage, distribution shift, conformal prediction, selective prediction

1. Introduction

Short-term network traffic forecasting is a core perception service in an intelligent transportation system (ITS). Forecasts of speed, flow, and occupancy support route guidance, ramp metering, signal coordination, congestion warning, incident management, emergency response, and the timing of traveler-information messages. A useful forecast must therefore do more than minimize an offline average: it must arrive within an operational latency budget, remain usable when field sensors fail, and expose enough uncertainty for a downstream controller to decide whether intervention is warranted. METR-LA and PEMS-BAY have enabled reproducible progress on this problem by representing a freeway system as multivariate time series attached to a sensor graph [1]. They remain the most common common-denominator benchmarks for comparing graph-based forecasting methods, despite their age and highway-centric scope.

The dominant modeling trajectory has been to improve spatiotemporal feature extraction. DCRNN models directed traffic diffusion inside a recurrent encoder–decoder [1]; STGCN couples spectral graph operations with gated temporal convolutions [2]; and Graph WaveNet learns an adaptive adjacency together with dilated temporal filters [3]. Later systems learn node-specific parameters [4], discrete or dynamic graphs [5, 6], synchronous spatiotemporal neighborhoods [7], propagation delays [8], or memory-conditioned meta-graphs [9]. Attention and transformer designs extend the receptive field [10, 11, 12], whereas STID and FC-GAGA show that simple identity embeddings or fully connected gated blocks can be surprisingly competitive [13, 14]. The lesson is important for credible novelty: additional architectural complexity is not, by itself, evidence of a better deployable forecaster.

Most benchmark tables nevertheless assume that the input graph is fully observed and that the test distribution is represented by a single clean chronological block. This assumption is fragile in practice. Loop detectors and roadside units may be unavailable because of communication loss, maintenance, power interruption, calibration drift, or shared infrastructure failure. Such

missingness is rarely independent. A communications outage can remove a contiguous corridor; a gateway failure can remove high-centrality sensors; and intermittent hardware can create temporal bursts. At the same time, traffic can move from free flow to congestion, recover after an incident, or exhibit volatility not represented in training. Graph message passing can amplify these failures because a corrupted or absent node influences its neighbors. Work on tensor completion and graph-tensor imputation demonstrates that topology helps recover missing network observations [15, 16], while masked pretraining improves representation learning [17, 18]. Yet imputation accuracy or independent random masking does not answer how a forecaster behaves under simultaneous regime shift and spatially correlated outage.

Average point error is also an incomplete operational signal. A low test MAE can coexist with large errors during a congestion transition or at a small set of safety-critical sensors. Likewise, 90% marginal prediction-interval coverage can hide serious undercoverage at particular nodes, horizons, or traffic regimes. Deep ensembles and probabilistic spatiotemporal models expose aleatoric or epistemic variation [19, 20], and conformal prediction can wrap a black-box forecaster with frequentist intervals under stated assumptions [21]. Online conformal methods respond to changing distributions [22, 23]. However, an interval alone does not tell an ITS whether a complete 12-step network forecast should be acted on. Selective prediction provides the missing decision layer: the system may retain a forecast when its estimated risk is acceptable and abstain when it is not. That choice is naturally evaluated through retained coverage, risk–coverage curves, and the failure rate among accepted forecasts, not only through interval width.

These observations motivate the joint question studied here: *when both the traffic regime and sensor-availability process differ from training, can a computationally compact forecaster deliver accurate point predictions, well-calibrated node–horizon intervals, and a leakage-free forecast-origin decision to act or abstain?* The question is deliberately stricter than robust point forecasting. It treats the output consumed by an ITS as one network-wide forecast tensor, acknowledges dependence among its node–horizon cells, and requires calibration and threshold selection to use only information available before the current target resolves. Recent traffic conformal approaches adapt intervals to topology or volatility [24, 25], and recent missing-sensor methods address cold-start or reliability-aware graph construction [26, 27]. Our 36-study evidence audit found no prior method that jointly benchmarks point prediction, subgroup interval reliability, and origin-level abstention across

independent, temporal-burst, contiguous-region, and high-centrality outages on both METR-LA and PEMS-BAY.

We address the gap with ShiftGuard. Its forecasting component concatenates speed and observation masks, learns a lightweight temporal representation, and combines static and adaptive spatial propagation. Training applies a curriculum of structured outages so that the model must use partial histories rather than learn only the clean sensor manifold. A probabilistic head supplies heterogeneity-aware scales. At deployment, node–horizon nonconformity scores are updated prequentially from resolved past origins, and interval width ranks whole forecast tensors. A disjoint construction/certification split among past origins prevents the selective threshold from being chosen and evaluated on the same outcomes. We do not claim a distribution-free guarantee under arbitrary temporal dependence; empirical prequential reliability and the assumptions of the binomial certificate are reported separately.

The experimental motivation is equally important. Clean accuracy alone would lead to an incorrect conclusion: ShiftGuard is statistically tied with the local STAEformer on METR-LA and is 0.027 mph worse on clean PEMS-BAY. Under the registered compound-shift protocol, however, the ordering changes. Across four outage families, five severities, two datasets, and five seeds, ShiftGuard reduces mean MAE relative to STAEformer by 0.347 mph on METR-LA and 0.546 mph on PEMS-BAY. At 30% severity it retains 86% and 98% of forecast origins while limiting empirical accepted-origin failure to 8% and 2%. It also uses about 7–14 times fewer parameters than STAEformer, depending on the dataset. These results position the work as a robustness and decision-reliability contribution, not as a universal clean-data state of the art.

This work makes four contributions:

1. We formulate traffic forecasting under compound temporal and measurement shift as a joint point, interval, and selective-decision task, and release a leakage-controlled protocol crossing traffic regimes with four graph-aware sensor-outage families and five severities.
2. We develop a compact mask-aware graph–temporal forecaster trained by structured-outage augmentation and coupled to heterogeneity-aware probabilistic scale estimation.
3. We introduce prequential node–horizon conformalization and an origin-level risk controller whose construction and certification samples are disjoint; reliability is evaluated using marginal and worst-group cover-

age, AURC, retention, and accepted-origin failure.

4. We provide a five-seed, 640-run comparison against classical, recurrent, graph-convolutional, adaptive-graph, identity-based, and transformer baselines, together with ablations, hierarchical paired inference, runtime, memory, and parameter analyses. Negative ablations are used to narrow rather than inflate the novelty claim.

The remainder of the article is organized as follows. Section 2 synthesizes deterministic graph forecasting, shift and missing-data robustness, probabilistic calibration, and selective prediction, and converts the survey into an explicit gap analysis. Section 3 formulates the compound-shift task. Section 4 presents the complete ShiftGuard architecture, learning objective, calibration procedure, and decision rule. Section 5 describes the datasets, splits, baselines, outage generators, metrics, and statistical protocol. Section 6 reports clean accuracy, comparison charts, robustness, calibration, ablation, and efficiency. Sections 7–9 discuss the evidence, limitations, future work, and conclusions.

2. Literature survey and gap analysis

2.1. From recurrent diffusion to adaptive graph learning

Early deep traffic forecasting established the now-standard decomposition into temporal dynamics and spatial interaction. DCRNN embeds a random-walk diffusion operator in a sequence-to-sequence recurrent architecture, reflecting the directed propagation of road conditions [1]. STGCN replaces recurrence with graph and gated temporal convolutions for efficient parallel training [2]. Graph WaveNet learns a latent adjacency and uses dilated causal convolutions, demonstrating that data-driven connectivity can complement or replace a distance graph [3]. These methods remain indispensable controls because many later gains depend on preprocessing, split conventions, and implementation details as much as on the proposed operator.

Subsequent work increases spatial adaptivity and heterogeneity. AGCRN generates node-specific parameters from learned embeddings [4]; GTS learns a discrete graph jointly with prediction [5]; and MTGNN learns directed relations among variables without requiring a fixed road graph [28]. STSGCN constructs localized synchronous spatiotemporal graphs rather than treating space and time as sequentially separable [7]. FC-GAGA uses fully connected graph gates and emphasizes computational efficiency [14]. D2STGNN

decomposes traffic into inherent and diffusion signals while learning a dynamic graph [6]; MegaCRN retrieves meta-graph information from a memory bank [9]; and PDFormer represents delayed propagation and long-range spatial dependence [8]. DSTAGNN similarly uses dynamic spatial-temporal awareness to model time-varying relations [11]. These advances improve representation power, but the observation mask is generally not an operational output and structured sensor failure is not the primary evaluation axis.

Attention-based architectures extend this progression. GMAN uses transform attention between encoded history and prediction horizons [10]. STAEformer shows that vanilla self-attention becomes a strong traffic forecaster when equipped with adaptive spatiotemporal embeddings [12]. Weaver factorizes full spatiotemporal attention by a Kronecker approximation and introduces signed valence relationships [29]. In parallel, STID demonstrates that spatial identity and temporal index embeddings can compete with substantially more complicated graph models [13]. The critical benchmark audit of Martin et al. further reports that carefully trained spatially unaware models are stronger than many published comparisons imply [30]. Consequently, our baseline suite includes persistence, ridge, linear, MLP, GRU, STGCN, DCRNN, Graph WaveNet, AGCRN, STID, and STAEformer under one pipeline, and parameter and latency measurements accompany error metrics.

2.2. Long context, pretraining, routing, and nonstationarity

Several approaches seek longer context or reusable representations rather than ever deeper graph stacks. STEP pretrains a scalable temporal encoder on long histories and supplies segment representations to a downstream STGNN [31]. STMAE masks graph-aware spatial tokens and temporal patches [17]; STD-MAE decouples spatial and temporal masked reconstruction [18]. These methods show that masking is a useful learning signal, but their objective is representation quality on the natural data distribution. ShiftGuard instead treats masks as explicit observations and generates failures whose spatial and temporal geometry matches deployment hazards.

Mixture and adaptation models address heterogeneity across contexts. TESTAM routes recurring and non-recurring conditions among specialized spatial experts [32]. CompFormer retrieves analogous historical representations at test time to compensate extreme traffic patterns [33]. Expand-and-Compress uses prompt pools for continual spatiotemporal forecasting

as streams and sensor networks evolve [34]. CAST-GNN learns causal invariances and dynamic associations to generalize from observed weekdays to unseen contexts [27]. These studies demonstrate that nonstationarity cannot be dismissed as random noise. Nevertheless, adapting a point predictor does not establish calibrated subgroup uncertainty, and online parameter tuning is conceptually different from deciding that a forecast tensor is unsafe to consume.

2.3. Missing observations, outages, and graph reliability

Missing traffic data has a long history in matrix and tensor completion. Regularized tensor factorization jointly imputes missing road states and learns latent temporal factors [15]. Graph-tensor neural networks unfold topology-aware completion algorithms for efficient network-wide imputation [16]. Heterogeneous graph models and availability-aware propagation likewise show that missing indicators should not be confused with genuine zero traffic. These approaches usually optimize reconstruction first, or couple imputation and forecasting under random missing ratios. A forecast model can therefore score well even if it has not been exposed to complete corridor or hub failures.

The distinction among failure geometries matters. Independent point loss tests local redundancy. A temporal burst tests whether the model can extrapolate when a sensor is silent for consecutive steps. A contiguous-region outage removes a spatial cluster and therefore challenges neighborhood aggregation. A hub outage targets sensors with high graph degree and can disproportionately interrupt message flow. SC-SAMG studies cold-start forecasting for unseen sensors through support-set adaptation [26], while recent reliability-oriented graphs incorporate sensor confidence or physical constraints. However, cold-start adaptation presumes a different information pattern from a sudden deployment outage, and a reliability weight is not itself evidence of robust decision quality. Our protocol therefore shares identical saved masks across models and crosses each failure geometry with pre-defined traffic regimes.

2.4. Probabilistic traffic forecasting and conformal calibration

Probabilistic forecasting supplements a conditional center with an estimate of dispersion or a predictive distribution. Bayesian graph neural networks, quantile regression, heteroscedastic likelihoods, and deep ensembles represent different combinations of aleatoric and epistemic uncertainty. A

systematic study by Wu et al. compares Bayesian and frequentist spatiotemporal uncertainty routes [19]; a later deep-ensemble framework combines simultaneous quantile regression with ensemble diversity [20]. These methods are informative but may be miscalibrated when their likelihood family or test distribution is misspecified.

Conformal prediction is attractive because it calibrates residual scores around an arbitrary base predictor. Conformal time-series forecasting extends the inductive framework to multi-horizon outputs under exchangeability conditions [21]. Gibbs and Candès address online prediction under arbitrary distribution shifts through locally adaptive updates [22], while decaying-step online conformal prediction improves behavior when a stable population quantile exists [23]. CAP studies false-coverage-rate control after adaptive selection [35]. These theoretical distinctions are essential: retrospective average coverage for an arbitrary sequence, marginal coverage under exchangeability, and prospective conditional coverage are not interchangeable claims.

Traffic-specific conformal models have begun to encode topology and volatility. Patil et al. combine availability-aware graph construction with adaptive intervals [24]. CSTH conditions interval scales on hypergraph density and traffic volatility and evaluates blocked-time calibration on METR-LA and PEMS-BAY [25]. A regime-dependent audit shows that aggregate interval coverage can conceal transition failures [36]. This motivates our node–horizon and regime reporting. The remaining gap is operational: none of these systems jointly evaluates structured graph outages and a risk-certified option to withhold an entire network forecast.

2.5. Selective prediction and operational decision making

Selective prediction couples a predictor with a selection function. Instead of forcing a low-confidence output into a controller, the system trades coverage for lower conditional risk. In traffic forecasting this trade-off must be defined carefully because one forecast origin contains hundreds or thousands of correlated node–horizon cells. Treating every cell as an independent trial would produce spuriously large calibration samples and unrealistically tight certificates. We therefore define operational failure using the mean absolute error of the complete forecast tensor and use its mean interval width as the selection score. The threshold is constructed and certified on disjoint past origins. CAP provides related theory for adaptively selected conformal outputs [35], but graph outages and network-wide forecast decisions are outside

Table 1: Taxonomy of representative literature and the unresolved joint task. "Struct." denotes explicitly structured sensor outages; "groups" denotes node, horizon, or regime reliability; and "abstain" denotes an operational selection decision rather than an interval alone.

Research line	Representative methods	Point forecast	Shift adaptation	Struct. outage	UQ/groups	Abstain
Diffusion/convolution	DCRNN, STGCN, Graph WaveNet	yes	no	no	no	no
Adaptive graph	AGCRN, GTS, D2STGNN, MegaCRN	yes	partial	no	no	no
Attention/transformer	GMAN, PDFormer, STAEformer, Weaver	yes	partial	no	no	no
Simple/efficient	STID, FC-GAGA	yes	no	no	no	no
Pretraining	STEP, STMAE, STD-MAE	yes	partial	masking proxy	no	no
Extreme/continual	CompFormer, Expand–Compress, CAST-GNN	yes	yes	no	no	no
Imputation	tensor factorization, graph-tensor networks	indirect	no	partial	no	no
Probabilistic traffic	Bayesian/ensemble STGNNs	yes	no	no	intervals/no	no
Conformal traffic	availability-aware CP, CSTH	yes	yes	limited	intervals/partial	no
Selective inference	CAP, online conformal prediction	generic	yes	no	generic	yes
ShiftGuard protocol	structured augmentation + prequential risk control	yes	yes	four families	intervals/groups	yes

its application setting.

2.6. Benchmarking practice and reproducibility

Traffic-forecasting comparisons are unusually sensitive to protocol choices. Published METR-LA and PEMS-BAY tables may differ in the raw-file revision, interpretation of zero values, chronological boundary, scaler fitting, horizon-wise aggregation, treatment of missing targets, and whether validation or test data influence graph construction. A model can also be reported from its best seed while another is averaged over seeds. The critical audit of Martin et al. demonstrates that these details can reverse conclusions about the value of spatial modeling [30]. Consequently, literature-reported numbers are useful for historical context but are not exchangeable with results from a new local pipeline.

Robustness evaluation introduces further degrees of freedom. Random masks must be shared across models; otherwise a method may receive an easier failed-sensor set. Severity may refer to cells, nodes, or complete temporal windows, which produce qualitatively different information loss. Calibration can be static, rolling, global, or group-specific. If a test residual helps select its own interval quantile or abstention threshold, apparent reliability is optimistic. Finally, reporting only the average across outage families can hide a method that fails catastrophically on a contiguous region or central hub. These concerns motivate our saved-mask design, family-wise artifacts, chronological prequential updates, and explicit separation of threshold construction from certification.

Efficiency is also part of reproducibility. Transformer accuracy may require a parameter, memory, and latency budget that is unavailable at a traffic control center or roadside device, while lightweight identity models

can provide a stronger practical reference [13, 12]. Parameter count alone is insufficient because recurrent sequential operations, dense attention, and graph kernels have different runtime behavior. We therefore measure profiler-observed operations, batch-one latency, batch-eight throughput, and peak allocated GPU memory on one disclosed platform. These measurements are not universal hardware rankings, but they make the local accuracy–reliability trade-off auditable.

Taken together, reproducibility requires more than releasing model code. It requires freezing data transforms, saving generated failures, distinguishing local reproductions from external numbers, retaining all confirmatory seeds, and regenerating tables from structured outputs. The present study treats this evaluation system as part of the contribution because a claim about reliable forecasting cannot be stronger than the protocol used to produce it.

2.7. Gap synthesis and research questions

Table 1 reveals four gaps. First, deterministic traffic forecasting and missing-data reconstruction are usually evaluated as separate tasks, leaving compound regime–outage shift under-specified. Second, probabilistic and conformal studies emphasize marginal coverage, which can hide unreliable nodes, horizons, and transition regimes. Third, few traffic systems convert uncertainty into an origin-level act/abstain decision with explicit risk and retention metrics. Fourth, complexity can obscure the true cause of improvement unless simple baselines, efficiency, and negative ablations are reported. We consequently ask: (RQ1) Does structured-outage training improve point robustness across both benchmarks and all four outage geometries? (RQ2) Does prequential group calibration improve worst-group reliability under drift? (RQ3) Can origin-level selection reduce accepted-forecast failure while retaining useful coverage? (RQ4) Which architectural or training components actually explain the gains, and what computational cost do they impose?

3. Problem formulation

3.1. Traffic network, observations, and prediction target

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)$ denote a directed traffic-sensor network, where $\mathcal{V} = \{v_1, \ldots, v_N\}$ is the sensor set, $\mathcal{E}$ contains spatial relations, and $A \in \mathbb{R}_+^{N \times N}$ is a weighted adjacency matrix. An edge weight may encode road-network proximity or empirical dependence between two sensors. Let $Z_t \in \mathbb{R}^N$ be

the conceptually complete vector of traffic speeds at five-minute instant t. Because measurements can be naturally absent or deliberately removed by a stress scenario, the model observes

$$X_t = M_t \odot Z_t, \qquad M_t \in \{0,1\}^N, \tag{1}$$

where $M_{t,i} = 1$ means that the speed at sensor i is available. In the implementation, masking is applied after standardization and the binary indicator is passed to the model; hence a missing value is distinguishable from a genuine zero-speed observation.

A forecast origin t uses a history of $P = 12$ observations, corresponding to the preceding 60 minutes,

$$\mathbf{X}_t = [X_{t-P+1}, \ldots, X_t] \in \mathbb{R}^{P\times N}, \qquad \mathbf{M}_t = [M_{t-P+1}, \ldots, M_t] \in \{0,1\}^{P\times N}, \tag{2}$$

together with calendar covariates C_t and graph $\mathcal{G}$. The target is the next $H = 12$ speed vectors,

$$\mathbf{Y}_t = [Z_{t+1}, \ldots, Z_{t+H}] \in \mathbb{R}^{H\times N}. \tag{3}$$

Any intrinsically unavailable target cell is excluded by a target-validity mask $O_t \in \{0,1\}^{H\times N}$. A forecasting system is therefore a mapping

$$f_\theta : (\mathbf{X}_t, \mathbf{M}_t, C_t, \mathcal{G}) \longmapsto (\widehat{\mathbf{Y}}_t, \mathbf{S}_t), \tag{4}$$

where $\widehat{\mathbf{Y}}_t$ is the point-forecast tensor and $\mathbf{S}_t > 0$ is a learned uncertainty scale for every horizon–sensor cell.

3.2. Compound regime and sensor-availability shift

Let $e_t \in \mathcal{E}_{\text{env}}$ describe the traffic environment at origin t, including recurrent congestion, free flow, transition periods, and atypical demand. The joint data-generating distribution is written as

$$(\mathbf{X}_t, \mathbf{M}_t, \mathbf{Y}_t, C_t) \sim \mathcal{P}_{e_t}. \tag{5}$$

Standard empirical-risk minimization implicitly expects future origins to resemble the training mixture $\mathcal{P}_{\text{tr}}$. The deployment setting considered here permits both the traffic mechanism and the observation mechanism to change:

$$\mathcal{P}_{\text{te}}(\mathbf{Y}_t \mid \mathbf{X}_t, C_t, \mathcal{G}) \neq \mathcal{P}_{\text{tr}}(\mathbf{Y}_t \mid \mathbf{X}_t, C_t, \mathcal{G}), \quad \mathcal{P}_{\text{te}}(\mathbf{M}_t \mid e_t, \mathcal{G}) \neq \mathcal{P}_{\text{tr}}(\mathbf{M}_t \mid e_t, \mathcal{G}). \tag{6}$$

The first inequality represents regime shift; the second represents sensor outage shift. Their concurrence is a *compound shift*. Missingness is not assumed to be independent and identically distributed. Instead, an outage operator $\mathcal{T}_{\omega,s}$ transforms the intrinsic mask according to family ω and severity s:

$$\widetilde{\mathbf{M}}_t = \mathcal{T}_{\omega,s}(\mathbf{M}_t; \mathcal{G}), \qquad \omega \in \{\text{independent, burst, region, hub}\}. \tag{7}$$

Independent outages remove sensor–time cells, bursts remove temporally contiguous suffixes, region outages remove connected subgraphs, and hub outages remove high-centrality sensors. Thus the shift may destroy precisely the neighbors on which a graph forecaster relies. Severity $s \in [0, 1]$ controls the requested fraction of unavailable observations, subject to the geometry of the selected family.

3.3. Point and probabilistic forecasting objectives

For a single origin, masked mean absolute error is

$$\ell_{\mathrm{MAE}}(t) = \frac{\sum_{h=1}^{H} \sum_{i=1}^{N} O_{t,h,i} |Y_{t,h,i} - \widehat{Y}_{t,h,i}|}{\sum_{h=1}^{H} \sum_{i=1}^{N} O_{t,h,i}}. \tag{8}$$

The first objective is to minimize expected point risk not only for clean origins but across an explicit set $\mathcal{Q}$ of regime–outage distributions. This is expressed as a mixture risk and a worst-condition risk:

$$R_{\mathrm{avg}}(f) = \sum_{q \in \mathcal{Q}} w_q \mathbb{E}_q[\ell_{\mathrm{MAE}}], \qquad R_{\mathrm{worst}}(f) = \max_{q \in \mathcal{Q}} \mathbb{E}_q[\ell_{\mathrm{MAE}}], \tag{9}$$

where $w_q \geq 0$ and $\sum_q w_q = 1$. Reporting both quantities prevents a large gain on an easy outage from concealing failure on a correlated region or hub outage. RMSE and MAPE are retained as complementary evaluation measures, but MAE is the primary optimization and comparison metric because it is expressed in operationally interpretable miles per hour.

Point accuracy alone does not specify whether a forecast is trustworthy. From $\widehat{\mathbf{Y}}_t$ and $\mathbf{S}_t$, a calibration rule constructs intervals $[L_{t,h,i}, U_{t,h,i}]$. For nominal miscoverage α, the desired marginal constraint is

$$\Pr\{Y_{t,h,i} \in [L_{t,h,i}, U_{t,h,i}]\} \geq 1 - \alpha, \tag{10}$$

while interval efficiency is measured by mean width $W_t = (HN)^{-1} \sum_{h,i} (U_{t,h,i} - L_{t,h,i})$. Because aggregate coverage can hide unreliable nodes, horizons, or

regimes, coverage is also evaluated within groups $g \in \mathcal{G}_{\text{cal}}$. The reliability objective is to keep the worst-group coverage deficit

$$\Delta_{\text{wg}} = \max_{g \in \mathcal{G}_{\text{cal}}} \left[(1-\alpha) - \widehat{\text{Cov}}_g \right]_+ \tag{11}$$

small without obtaining coverage through uninformatively wide intervals.

3.4. Origin-level selective risk

Network forecasts are consumed jointly by downstream control systems. ShiftGuard therefore releases or withholds the complete $H \times N$ tensor at each origin rather than selecting isolated cells. Let $D_t = \delta(W_t) \in \{0, 1\}$ denote the decision, where $D_t = 1$ releases the forecast and $D_t = 0$ abstains. Define the realized origin error and operational failure event as

$$e_t = \ell_{\text{MAE}}(t), \qquad F_t = \mathbb{1}\{e_t > \eta\}, \tag{12}$$

with the preregistered tolerance $\eta = 5$ mph. Retention and selective failure risk are

$$\text{Ret}(\delta) = \Pr(D_t = 1), \qquad R_{\text{sel}}(\delta) = \Pr(F_t = 1 \mid D_t = 1). \tag{13}$$

The operational optimization problem is

$$\max_{\delta \in \mathcal{D}} \ \text{Ret}(\delta) \quad \text{subject to} \quad R_{\text{sel}}(\delta) \le \rho, \tag{14}$$

where $\rho = 0.10$. The constraint captures the asymmetric cost of deployment: abstaining reduces automation coverage, whereas releasing a severely erroneous network forecast can misdirect routing or control. Since R_{sel} is unknown, certification uses only resolved calibration origins and requires a simultaneous upper confidence bound no larger than ρ. If no candidate threshold satisfies the bound, the safe action is abstention.

3.5. Learning problem and leakage constraints

The complete task is to learn parameters θ, a prequential calibration map, and a decision rule δ that jointly: (i) minimize clean and compound-shift point risks in Eq. (9); (ii) produce informative intervals satisfying Eq. (10) with a small worst-group gap; and (iii) maximize retention under the selective-risk constraint in Eq. (14). These objectives are related but not interchangeable: a model can have low average MAE yet poor outage

robustness, nominal marginal coverage yet weak subgroup coverage, or high retention yet an unsafe conditional failure rate.

Chronology imposes the final constraint. Model parameters and normalization statistics use the training split only; hyperparameters and initial calibration use the validation split; and a prediction at origin t may use only inputs available by t and outcomes resolved strictly before t. The target $\mathbf{Y}_t$ cannot influence its own interval, calibration quantile, or release decision. This filtration-respecting formulation distinguishes the proposed operational problem from a retrospective benchmark in which all test residuals are calibrated or ranked jointly.

4. Method

4.1. System architecture and data flow

ShiftGuard is organized into three explicitly separated layers: a reliability-aware point forecaster, a prequential uncertainty-calibration layer, and an operational risk controller. Figure 1 shows the complete training and deployment architecture, including the information permitted to cross each boundary. The separation is important because a low-error neural predictor does not automatically provide calibrated intervals, and a calibrated interval does not by itself specify when an ITS should act on or reject a complete network forecast.

The forward path begins at block 1 with the latest $P \times N$ speed window, its intrinsic observation mask, calendar variables, and the sensor graph. During training, the structured-outage generator in block T1 samples an outage family and severity and combines the resulting stress mask with the intrinsic mask. At deployment this branch is disabled: only genuinely available readings are used. Block 2 standardizes observed speeds with training-set statistics, zeros unavailable values only after standardization, and concatenates the binary mask so that missing readings cannot be confused with physical zero speed. Block 3 then combines a node-shared temporal encoder with node identity, time-of-day, and day-of-week embeddings to produce one latent state per sensor.

Spatial inference occurs in block 4. A distance- or correlation-derived static support and a learned adaptive support are fused, while each sender is weighted by the fraction of its history that remains observable. Three gated graph blocks exchange neighbor messages and retain a residual local path; therefore, the architecture can reduce reliance on graph messages when the

most informative neighbors have failed. In parallel, block 5 summarizes the current origin using mean speed, speed dispersion, temporal variation, and mean sensor availability. A compact router maps these statistics to mixture weights for four output experts. Block 6 combines their residual location and positive scale estimates with persistence and a learned history skip, producing the $H \times N$ median forecast $\widehat{\mathbf{Y}}_t$ and raw scale $\mathbf{S}_t$.

The magenta training path explains how model parameters are learned. Observed target cells enter a masked Laplace negative log-likelihood in block T2; missing targets never contribute to the loss. Gradients update the temporal encoder, graph supports, router, and probabilistic head jointly. The structured-outage curriculum changes the training distribution rather than repairing inputs with a separate imputation model. Consequently, the predictor learns to use the mask, surviving local history, and available graph context under independent, burst, and whole-node failures.

The lower deployment path is frozen with respect to neural parameters. Before origin t is issued, block D1 contains only validation residuals and residuals from test origins whose targets have already resolved. Block D2 converts these past normalized errors into pooled and node–horizon conformal quantiles and forms calibrated intervals around the current point forecast. Block D3 reduces the interval tensor to an origin-level uncertainty score, compares it with a threshold certified on disjoint past origins, and either releases the complete network forecast or abstains. After the future target becomes observable, the new residual is appended to D1 for subsequent origins, as shown by the dashed feedback arrow. The target of the current origin is therefore unavailable to the forecaster, calibrator, and decision rule at decision time. This ordered data flow is the main leakage-control invariant and enables point accuracy, interval reliability, and selective risk to be audited separately.

4.2. Reliability-conditioned encoding

For a mini-batch of size B, the raw model input has shape $B \times P \times N \times F$, where the first feature is standardized speed and the remaining channels include the intrinsic observation indicator, time of day, and day of week. A stress mask M_t is multiplied by the intrinsic indicator, so naturally absent measurements remain absent. Unobserved speeds are set to zero only after standardization and are never confused with true zero speed because the binary mask is supplied separately. For node i, the encoder input is

$$q_{t,i} = \mathrm{vec}\left([X_{t-P+1:t,i} \odot M_{t,i}, M_{t,i}]\right) \in \mathbb{R}^{2P}. \tag{15}$$

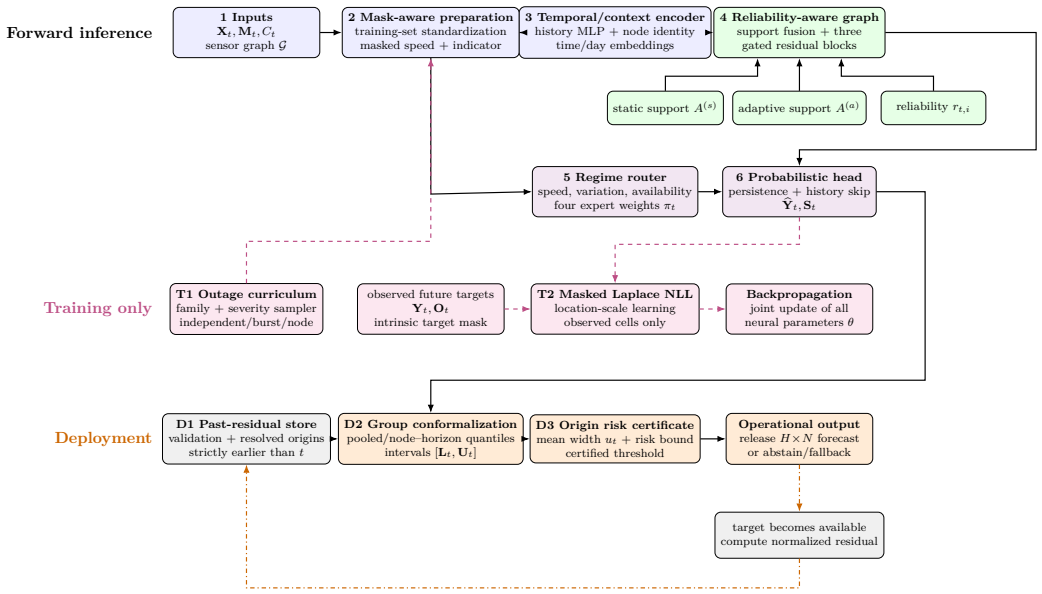


Figure 1: Detailed ShiftGuard architecture and leakage-controlled data flow. The blue/green/violet forward path produces node–horizon locations and scales; the magenta path applies structured-outage training and masked probabilistic loss; and the orange deployment path performs prequential conformal calibration and origin-level release or abstention. Solid arrows denote current-origin information, dashed arrows denote training-only operations, and dash-dotted arrows denote feedback available only after a target has resolved.

A two-layer GELU multilayer perceptron maps $q_{t,i}$ to a $d = 64$ temporal state. Learned node identity, five-minute time-of-day, and day-of-week embeddings are added to this state. This direct construction is deliberately small: it preserves the strong identity-based inductive bias documented by STID [13] without requiring full spatiotemporal self-attention.

The average availability of node i in the input window is

$$r_{t,i} = P^{-1} \sum_{p=1}^{P} M_{t-p+1,i}. \tag{16}$$

The static support $A^{(s)}$ is the distance-derived benchmark adjacency with self-loops and row normalization. The adaptive support is

$$A^{(a)} = \operatorname{softmax}\big(\operatorname{ReLU}(E_s E_d^{\top})\big), \tag{17}$$

where $E_s, E_d \in \mathbb{R}^{N \times 16}$ are learned source and destination embeddings. When

both supports are enabled, their equal-weight mixture is sender-weighted and normalized:

$$\widetilde{A}_{t,ij} = \frac{\frac{1}{2}(A^{(s)}_{ij} + A^{(a)}_{ij})r_{t,j}}{\sum_k \frac{1}{2}(A^{(s)}_{ik} + A^{(a)}_{ik})r_{t,k} + \epsilon}. \tag{18}$$

Thus a sender with no usable history contributes little to its neighbors. Each of three graph blocks computes a linear message $m_i = \sum_j \widetilde{A}_{t,ij} W_m z_j$ and a gate $g_i = \sigma(W_g[z_i, m_i, r_i])$. The update is a normalized residual mixture of m_i and z_i. Residual connections preserve local information when the graph is unreliable, while dropout regularizes the exchanged messages.

4.3. Regime routing and probabilistic head

Four global statistics summarize the current origin: mean speed, speed standard deviation, mean absolute first difference, and mean sensor availability. A two-layer router converts them to weights $\pi_t = \mathrm{softmax}(f_{\mathrm{route}}(c_t))$ over $K = 4$ compact output experts. Expert k maps every node state to H median residuals and H positive scales. The combined outputs are

$$\hat{Y}_t = X_{t,\mathrm{last}} + W_{\mathrm{skip}} X_{t-P+1:t} + \sum_{k=1}^{K} \pi_{t,k} \mu_{t,k}, \tag{19}$$

$$S_t = \sum_{k=1}^{K} \pi_{t,k} \left[\mathrm{softplus}(s_{t,k}) + 10^{-4}\right]. \tag{20}$$

The last-observation term makes persistence an easy initial solution, while the zero-initialized history projection learns a direct multi-horizon correction. The model has 89,012 parameters for METR-LA and 100,340 for PEMS-BAY. The router is retained in the evaluated full system for preregistration fidelity, but the ablation results later show that it is not the source of the robustness gain.

The location and scale are trained with the Laplace negative log likelihood over observed target cells,

$$\mathcal{L}_{\mathrm{Lap}} = \frac{1}{|\Omega|} \sum_{(h,i)\in\Omega} \left(\frac{|Y_{t,h,i} - \hat{Y}_{t,h,i}|}{S_{t,h,i}} + \log(2S_{t,h,i}) \right), \tag{21}$$

where Ω excludes unavailable ground truth. AdamW, gradient clipping, and validation early stopping optimize this objective. The probabilistic scale is not treated as calibrated by construction; it becomes a normalization factor for the conformal residual score.

4.4. Structured-outage curriculum

At every training batch, a seeded generator samples an outage family and severity. Independent loss removes individual sensor–time cells. Burst loss chooses sensor–window pairs and masks a suffix of each selected history. Region loss selects a random seed node and grows a connected failed set using the strongest unseen graph neighbor. Hub loss removes the nodes with largest combined in/out degree. The confirmatory test masks are generated once, stored, and reused by every model and seed, so differences cannot arise from different failure samples. The training curriculum includes independent, burst, and whole-node masks; region and hub tests assess whether this augmentation generalizes to unseen correlated geometry. The no-augmentation and independent-only variants directly test this mechanism.

4.5. Conformal risk control

For a resolved calibration origin u, node–horizon nonconformity is

$$R_{u,h,i} = \frac{|Y_{u,h,i} - \hat{Y}_{u,h,i}|}{S_{u,h,i}}. \tag{22}$$

Before predicting origin t, the rolling calibrator contains only $\{R_u : u < t\}$. It estimates the finite-sample conformal quantile separately for every (h, i) group and shrinks small groups toward the pooled quantile. If $n_{h,i}$ usable scores occur in the rolling window and τ is the shrinkage constant, the effective radius is based on $w_{h,i}q_{h,i}+(1-w_{h,i})q_{\text{global}}$, with $w_{h,i} = n_{h,i}/(n_{h,i}+\tau)$. The interval is

$$[L_{t,h,i}, U_{t,h,i}] = [\hat{Y}_{t,h,i} - q_{t,h,i}S_{t,h,i}, \hat{Y}_{t,h,i} + q_{t,h,i}S_{t,h,i}]. \tag{23}$$

Rolling updates allow the scale to react after distribution changes, but they do not convert dependent traffic into an exchangeable sample. We therefore report marginal, worst-node–horizon, and worst-regime empirical coverage and avoid a prospective conditional-coverage claim.

Selection operates at the forecast-origin level. Define realized operational error and uncertainty score as

$$e_t = (HN)^{-1} \sum_{h,i} |Y_{t,h,i} - \hat{Y}_{t,h,i}|, \qquad u_t = (HN)^{-1} \sum_{h,i} (U_{t,h,i} - L_{t,h,i}). \tag{24}$$

Candidate thresholds are quantiles of u_t from even-indexed past origins. For each threshold, odd-indexed past origins estimate the Bernoulli failure indicator $\mathbb{1}\{e_t > 5 \text{ mph}\}$ among accepted forecasts. The controller chooses the

largest retained fraction whose Bonferroni-corrected Clopper–Pearson upper confidence bound is at most the registered 10% risk limit at familywise $\delta = 0.05$. If no candidate is certified, it abstains. Sorting all origins by u_t also yields the risk–coverage curve and its area (AURC); lower AURC means that uncertainty ranks harmful forecasts more effectively.

At deployment, the sequence is therefore: (1) ingest the latest speed and availability window; (2) compute point and scale tensors; (3) form intervals from past residuals; (4) compare mean width with the certified threshold; (5) release or withhold the full tensor; and (6) after its target resolves, append the new residual to the rolling calibration store. This ordered pipeline is the central leakage-control invariant of the system.

5. Experimental design

5.1. Datasets and leakage control

METR-LA. The Los Angeles benchmark contains speed readings from 207 loop detectors on metropolitan-area freeways. The checksum-verified file used here spans 1 March to 27 June 2012 and contains 34,272 five-minute timestamps, equivalent to 119 days. After interpreting non-finite and non-positive readings as unavailable, 8.109% of sensor–time cells are missing. The observed mean is 58.460 mph and the standard deviation is 13.024 mph. METR-LA contains a meaningful mixture of operating conditions: 4.35% of valid readings are below 25 mph, 72.60% exceed 55 mph, and 14.09% of consecutive sensor readings change by more than 5 mph. Its pooled autocorrelation is 0.942 at one five-minute lag, 0.715 at one hour, 0.627 at one day, and 0.709 at one week. These statistics indicate strong persistence and periodicity but also substantially more native missingness and volatility than PEMS-BAY.

PEMS-BAY. The Bay Area benchmark contains 325 freeway sensors and 52,116 five-minute timestamps from 1 January to 30 June 2017, or 181 days. Its native missing fraction after the same validity rule is only 0.003%. Mean speed is 62.621 mph with a 9.588 mph standard deviation. Readings below 25 mph account for 1.92%, while 90.32% exceed 55 mph; only 2.30% of consecutive sensor measurements change by more than 5 mph. Autocorrelation is 0.983 at five minutes, 0.731 at one hour, 0.681 at one day, and 0.780 at one week. PEMS-BAY is therefore larger, cleaner, and more strongly persistent. Reporting both datasets is important because robustness that holds only on METR-LA could be an artifact of its existing missingness, while

Table 2: Dataset diagnostics computed from checksum-verified raw files. Speed is in miles per hour; correlations use observed values at 5-minute and one-day lags.

Dataset	Timestamps	Sensors	Missing (pct.)	Mean speed	Std. speed	Lag-1 corr.	Lag-288 corr.
METR-LA	34272	207	8.109	58.460	13.024	0.942	0.627
PEMS-BAY	52116	325	0.003	62.621	9.588	0.983	0.681

robustness that holds only on PEMS-BAY might not transfer to a noisier network.

Graph and feature construction. To avoid reliance on incompatible external graph files, the static support is reconstructed from the training input only. Pairwise sensor correlations are computed over training histories; for every node, the 12 strongest non-negative correlations are retained. The same support is supplied to all graph-based models and to the region/hub outage generators. Each window contains standardized speed, a native-availability indicator, normalized five-minute time of day, and normalized day of week. Only the speed channel is forecast. This correlation graph is a reproducible proxy, not a claim to recover true road causality, and its limitations are discussed in Section 7.

Chronological splits and windows. Raw timestamps are partitioned before windowing into 70% training, 10% validation, and 20% test segments. For METR-LA this produces 23,967 training, 3,416 validation, and 6,844 test forecast origins; for PEMS-BAY it produces 36,458, 5,200, and 10,413 origins. Twelve historical samples (60 minutes) directly predict the next twelve samples (60 minutes). Validation and test inputs may use the immediately preceding 60 minutes as context, but every target lies inside its own split. Consequently, a target never crosses backward into the training or validation block.

Missingness and normalization. Per-sensor mean and standard deviation are fitted only on observed training values. Missing input speed is replaced by the training mean before standardization, becoming zero in normalized space, and the mask channel preserves its unavailable status. Missing targets are excluded from loss and metrics. All reported MAE and RMSE values are inverse transformed to mph. Test data are never used for normalization, early stopping, hyperparameter selection, conformal initialization, or selection-threshold construction. Dataset diagnostics in Table 2 are generated directly from the raw files rather than copied from secondary papers.

Table 3: Literature-reported point accuracy for context only. Protocol and implementation differences prohibit direct significance comparisons with locally reproduced results.

Dataset	Model	MAE at 15 min	MAE at 30 min	MAE at 60 min	Average MAE
METR-LA	DCRNN	2.77	3.15	3.60	–
METR-LA	Graph WaveNet	2.69	3.08	3.51	–
METR-LA	STAEformer	2.65	2.97	3.34	–
METR-LA	TESTAM	–	–	–	2.93
PEMS-BAY	DCRNN	1.38	1.74	2.07	–
PEMS-BAY	Graph WaveNet	1.30	1.63	1.99	–
PEMS-BAY	STAEformer	1.31	1.62	1.88	–
PEMS-BAY	TESTAM	–	–	–	1.53

5.2. Baselines

Locally reproduced baselines include historical average, persistence, ridge AR, linear, MLP/GRU, STGCN, DCRNN, Graph WaveNet, AGCRN, STID, and a faithful STAEformer configuration. Compatible open implementations of recent methods are added after preprocessing and license audits. Reported SOTA values appear in a separate contextual table and are never used for significance claims. The recurrent and graph baselines are protocol-matched compact reproductions, not claims of bitwise equivalence to their authors' code; a model-by-model deviation audit accompanies the repository.

5.3. Stress tests and metrics

We cross clean, independent, temporal-burst, contiguous-region, and high-degree hub outages at 10–50% severity with free-flow, congested, stable intermediate, and transition regimes. Metrics include horizon-specific and average MAE, RMSE, MAPE; degradation and worst-regime MAE; coverage, width, Winkler score and worst-group coverage; AURC and selective MAE; parameters, FLOPs, latency, peak memory, and training time. Headline results use at least five seeds and paired bootstrap confidence intervals.

The confirmatory environment is captured automatically, including Python and package versions, CUDA build, GPU, and driver. Runs use PyTorch 2.13.0 with CUDA 13.0 on an NVIDIA GeForce RTX 5060 Laptop GPU; the generated environment manifest is retained with the result artifacts.

Table 4: Locally reproduced canonical-test results. Empty uncertainty cells indicate that only one deterministic run is applicable or currently complete.

Dataset	Model	MAE	SD	RMSE	MAPE (pct.)	Parameters	Train (s)	Seeds
METR-LA	STAEformer	3.032	0.008	6.178	8.362	1258932	4554.221	5
METR-LA	ShiftGuard	3.033	0.005	6.087	8.511	89012	357.521	5
METR-LA	DCRNN	3.247	0.010	6.380	9.133	63372	1002.598	5
METR-LA	STID	3.273	0.015	6.603	9.558	41996	90.004	5
METR-LA	Graph WaveNet	3.285	0.008	6.451	9.205	22744	749.354	5
METR-LA	AGCRN	3.343	0.010	6.576	9.494	16764	492.095	5
METR-LA	GRU	3.761	0.002	7.628	10.878	13644	464.916	5
METR-LA	MLP	3.798	0.002	7.662	11.032	5772	68.023	5
METR-LA	STGCN	3.899	0.013	7.187	11.365	17548	374.899	5
METR-LA	Linear	3.982	0.001	7.728	11.589	156	48.049	5
METR-LA	Historical average	4.189	–	7.856	13.044	0	0.000	1
METR-LA	Persistence	4.244	–	8.154	11.052	0	0.000	1
METR-LA	Ridge AR	4.294	–	7.595	12.727	324	0.093	1
PEMS-BAY	STAEformer	1.577	0.006	3.650	3.613	1372212	13896.209	5
PEMS-BAY	ShiftGuard	1.604	0.003	3.591	3.691	100340	860.822	5
PEMS-BAY	STID	1.664	0.004	3.739	3.822	49548	199.245	5
PEMS-BAY	DCRNN	1.683	0.005	3.779	3.900	63372	2888.655	5
PEMS-BAY	Graph WaveNet	1.720	0.005	3.770	3.934	25104	1830.569	5
PEMS-BAY	AGCRN	1.731	0.004	3.849	4.016	18652	1338.461	5
PEMS-BAY	GRU	1.955	0.001	4.619	4.566	13644	1081.630	5
PEMS-BAY	MLP	1.969	0.004	4.648	4.569	5772	169.266	5
PEMS-BAY	Linear	2.130	0.001	4.955	4.808	156	71.894	5
PEMS-BAY	Persistence	2.171	–	5.125	4.671	0	0.000	1
PEMS-BAY	Ridge AR	2.302	–	4.831	5.371	324	0.086	1
PEMS-BAY	STGCN	2.506	0.005	4.873	5.918	17548	903.628	5
PEMS-BAY	Historical average	2.630	–	5.228	6.176	0	0.000	1

6. Results

6.1. Canonical accuracy

Table 4 is generated exclusively from local experiment artifacts. A stochastic model enters the confirmatory table only after all five registered seeds complete; partial runs are omitted.

Figure 2 makes the clean-data ranking explicit. On METR-LA, ShiftGuard and STAEformer differ by only 0.001 mph in pooled MAE and both improve on the next group of graph baselines. On PEMS-BAY, STAEformer leads ShiftGuard by 0.027 mph; STID and DCRNN remain close behind. ShiftGuard has the lower RMSE of the two leading methods on both datasets (6.087 versus 6.178 and 3.591 versus 3.650), suggesting fewer large residuals even where its mean absolute error is slightly higher. This observation is descriptive; the paired origin-level analysis below determines inferential claims.

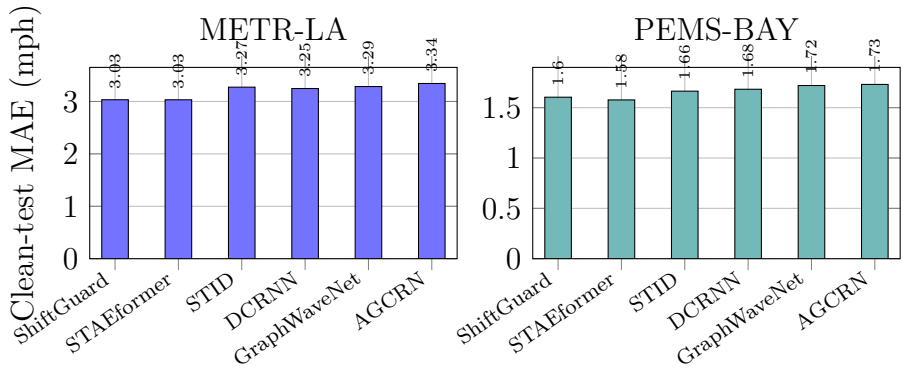


Figure 2: Clean-test comparison chart for the strongest locally reproduced neural baselines. ShiftGuard is effectively tied with STAEformer on METR-LA but does not achieve clean-data SOTA on PEMS-BAY.

Paired inference uses non-overlapping daily blocks of forecast origins. The reported confidence interval is a crossed seed–day paired-bootstrap interval for $\Delta\text{MAE} = \text{MAE}_{\text{ShiftGuard}} - \text{MAE}_{\text{comparator}}$; both seeds and chronological daily blocks are resampled. Wilcoxon tests operate on paired daily-block differences averaged over seeds and are Holm corrected across the displayed comparisons.

For METR-LA, the completed five-seed comparison does not establish clean-error superiority over the local STAEformer implementation: the origin-weighted MAE difference is -0.0007 mph (crossed-bootstrap 95% interval $[-0.0308, 0.0221]$ mph; Holm-adjusted $p = 0.944$). This statistical estimand weights forecast origins equally, whereas the canonical table pools observed node–horizon errors. Failure to detect a difference is not proof of equivalence. The proposed reliability contribution therefore requires independent support from the registered outage and selective-prediction experiments. On PEMS-BAY, STAEformer is better on the clean test: the origin-weighted difference is 0.0269 mph, with a 95% interval $[0.0155, 0.0374]$ and Holm-adjusted $p = 4.93 \times 10^{-6}$. Thus ShiftGuard is not presented as a new clean-data point-forecasting SOTA.

6.2. Compound-shift robustness and calibration

Table 6 averages the registered 30% non-clean outage families; all family-by-rate results remain available in machine-readable artifacts.

Across independent, burst, region, and hub outages at 10, 20, 30, 40, and

Table 5: Paired daily-block statistical comparison. Negative ΔMAE favors ShiftGuard.

Dataset	Comparator	Delta MAE	CI low	CI high	Improvement (pct.)	Holm p
METR-LA	agcrn	-0.3160	-0.3607	-0.2801	9.3884	< 0.0001
METR-LA	dcrnn	-0.2188	-0.2621	-0.1977	6.6928	< 0.0001
METR-LA	graphwavenet	-0.2559	-0.2867	-0.2267	7.7398	< 0.0001
METR-LA	gru	-0.7292	-0.8130	-0.6167	19.2941	< 0.0001
METR-LA	historical average	-1.1510	-1.3156	-1.0603	27.3965	< 0.0001
METR-LA	linear	-0.9513	-1.0477	-0.8318	23.7737	< 0.0001
METR-LA	mlp	-0.7707	-0.8650	-0.6721	20.1695	< 0.0001
METR-LA	persistence	-1.2139	-1.2951	-1.1067	28.4667	< 0.0001
METR-LA	ridge ar	-1.2623	-1.3503	-1.1482	29.2695	< 0.0001
METR-LA	staeformer	-0.0007	-0.0314	0.0216	0.0228	0.9441
METR-LA	stgcn	-0.8670	-0.9338	-0.8072	22.1324	< 0.0001
METR-LA	stid	-0.2431	-0.3038	-0.2065	7.3817	< 0.0001
PEMS-BAY	agcrn	-0.1264	-0.1455	-0.1036	7.3042	< 0.0001
PEMS-BAY	dcrnn	-0.0785	-0.0943	-0.0600	4.6670	< 0.0001
PEMS-BAY	graphwavenet	-0.1155	-0.1301	-0.0972	6.7159	< 0.0001
PEMS-BAY	gru	-0.3509	-0.4114	-0.2761	17.9473	< 0.0001
PEMS-BAY	historical average	-1.0262	-1.1972	-0.8952	39.0150	< 0.0001
PEMS-BAY	linear	-0.5258	-0.6103	-0.4193	24.6876	< 0.0001
PEMS-BAY	mlp	-0.3650	-0.4277	-0.2857	18.5373	< 0.0001
PEMS-BAY	persistence	-0.5664	-0.6506	-0.4624	26.0971	< 0.0001
PEMS-BAY	ridge ar	-0.6982	-0.7884	-0.5839	30.3263	< 0.0001
PEMS-BAY	staeformer	0.0269	0.0158	0.0376	-1.7064	< 0.0001
PEMS-BAY	stgcn	-0.9017	-0.9601	-0.8289	35.9857	< 0.0001
PEMS-BAY	stid	-0.0599	-0.0868	-0.0400	3.6001	< 0.0001

50%, the mean ShiftGuard–STAEformer MAE difference is -0.347 mph on METR-LA (crossed seed–scenario 95% interval $[-0.465, -0.234]$) and -0.546 mph on PEMS-BAY ($[-0.682, -0.411]$). Corresponding AURC differences are -1.46 and -1.26. At 30%, worst node–horizon coverage is 0.831 versus 0.816 on METR-LA and 0.851 versus 0.834 on PEMS-BAY; worst-regime coverage is 0.882 versus 0.809 and 0.887 versus 0.701. Seed-level Wilcoxon tests cannot reject after ten-comparison Holm correction ($p_{\text{Holm}} = 0.625$), whose attainable resolution is limited by five seeds; we therefore report the hierarchical intervals and effects but do not claim adjusted rank-test significance.

Severity sensitivity provides stronger evidence than a single 30% slice. On METR-LA, the ShiftGuard advantage over STAEformer grows from 0.133 mph at 10% outage to 0.521 mph at 50%; on PEMS-BAY it grows from 0.224 to 0.866 mph. Relative to STAEformer, the 50%-severity reductions are approximately 12.3% and 28.5%. The curves remain ordered at every

Table 6: Compound-shift point, interval, and selective reliability. PI is the 90% prediction interval; failure denotes accepted forecasts exceeding the registered 5 mph error tolerance. Worst node–horizon (N–H) and regime coverage are reported for ShiftGuard and its strongest local comparator.

Dataset	Model	MAE	PI coverage	Worst N-H cov.	Worst regime cov.	AURC	Retained	Failure rate	Seeds
METR-LA	AGCRN	4.729	0.891	–	–	4.950	0.000	–	5
METR-LA	DCRNN	4.629	0.891	–	–	4.791	0.000	–	5
METR-LA	Graph WaveNet	4.895	0.890	–	–	5.125	0.000	–	5
METR-LA	GRU	4.919	0.893	–	–	5.181	0.000	–	5
METR-LA	Linear	5.109	0.894	–	–	5.356	0.000	–	5
METR-LA	MLP	5.054	0.893	–	–	5.301	0.000	–	5
METR-LA	Persistence	5.478	0.895	–	–	5.703	0.000	–	5
METR-LA	ShiftGuard	3.399	0.898	0.831	0.882	2.400	0.864	0.078	5
METR-LA	STAEformer	3.760	0.894	0.816	0.809	3.873	0.055	0.235	5
METR-LA	STGCN	5.309	0.893	–	–	5.536	0.000	–	5
METR-LA	STID	4.092	0.894	–	–	4.256	0.000	–	5
PEMS-BAY	AGCRN	3.073	0.890	–	–	2.952	0.174	0.115	5
PEMS-BAY	DCRNN	2.956	0.889	–	–	2.844	0.183	0.109	5
PEMS-BAY	Graph WaveNet	2.928	0.889	–	–	2.824	0.203	0.102	5
PEMS-BAY	GRU	3.109	0.893	–	–	3.074	0.167	0.126	5
PEMS-BAY	Linear	3.199	0.894	–	–	3.221	0.077	0.203	5
PEMS-BAY	MLP	3.135	0.893	–	–	3.105	0.167	0.127	5
PEMS-BAY	Persistence	3.283	0.894	–	–	3.268	0.001	0.267	5
PEMS-BAY	ShiftGuard	1.948	0.898	0.851	0.887	1.130	0.978	0.020	5
PEMS-BAY	STAEformer	2.516	0.890	0.834	0.701	2.402	0.442	0.085	5
PEMS-BAY	STGCN	3.857	0.893	–	–	3.752	0.073	0.221	5
PEMS-BAY	STID	2.649	0.891	–	–	2.544	0.209	0.099	5

tested severity, so the aggregate effect is not driven by one operating point. STID is consistently less robust than both leading models, showing that a compact identity-based architecture alone is insufficient; exposure to structured partial observations is the critical distinction.

Failure geometry further qualifies the result. At 30% severity, ShiftGuard reduces STAEformer MAE by 0.382–0.523 mph on METR-LA for independent, burst, and hub outages, but by only 0.116 mph for a contiguous region. On PEMS-BAY the reductions range from 0.431 mph for region loss to 0.792 mph for hub loss. The region case removes an internally coherent subgraph and leaves fewer informative boundaries from which to infer missing states. Hub loss instead disrupts STAEformer's global spatial interaction more strongly, while ShiftGuard can fall back on its local residual path and training-time node outages.

6.3. Selective forecasting, ablation, and efficiency

At 30% severity, prequential risk control reduces accepted-origin failure on METR-LA from 15% without abstention to 8%, while retaining 86% of forecasts. On PEMS-BAY it reduces failure from 4% to 2% at 98% retention. Static calibration fails under drift on METR-LA (15% failure despite full re-

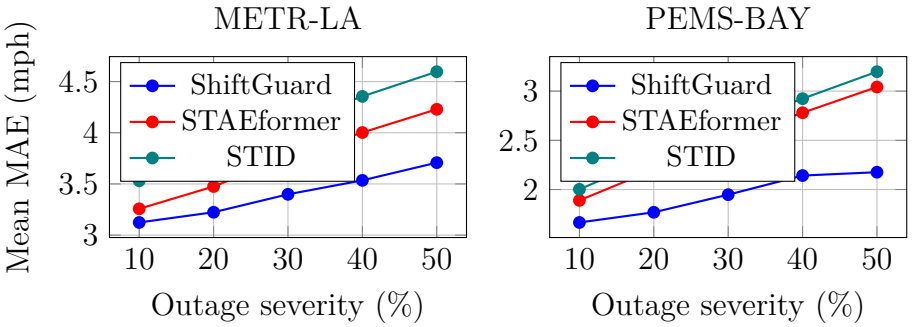


Figure 3: MAE averaged over independent, burst, region, and hub outages and five seeds. ShiftGuard's advantage increases with outage severity.

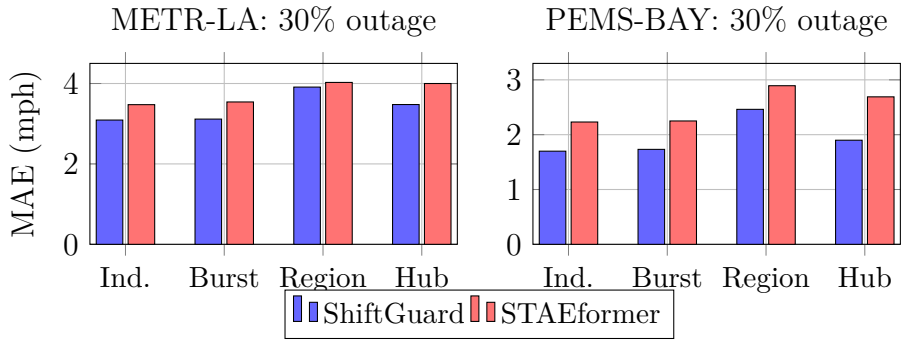


Figure 4: Outage-family comparison at 30% severity, averaged over five seeds (Ind. = independent loss). Contiguous-region failure is the hardest condition for ShiftGuard, whereas hub failure particularly harms STAEformer on PEMS-BAY.

tention), supporting online updates. Table 7 reports all registered calibration controls.

The calibration controls reveal a trade-off rather than a uniformly dominant update. Global-prequential and group-prequential ShiftGuard have nearly identical marginal coverage and operational results at the displayed precision. Static group calibration performs poorly on METR-LA: it retains 99.6% but fails on 15.2% of accepted origins. On the more stationary PEMS-BAY, the static variant achieves 1.5% failure at 95.0% retention, whereas prequential grouping achieves 2.0% failure at 97.8% retention. The value of rolling adaptation is therefore strongest on the dataset with more native missingness and transition activity; static calibration is not intrinsically inferior when the environment is stable.

Table 7: Calibration and selection controls at 30% outage severity, averaged over non-clean families and five seeds.

Dataset	Calibration	Selection	PI coverage	Retained	Failure rate
METR-LA	global-prequential	risk	0.898	0.864	0.079
METR-LA	group-prequential	all	0.898	1.000	0.154
METR-LA	group-prequential	risk	0.898	0.864	0.078
METR-LA	group-static	risk	0.893	0.996	0.152
PEMS-BAY	global-prequential	risk	0.898	0.977	0.020
PEMS-BAY	group-prequential	all	0.898	1.000	0.037
PEMS-BAY	group-prequential	risk	0.898	0.978	0.020
PEMS-BAY	group-static	risk	0.884	0.950	0.015

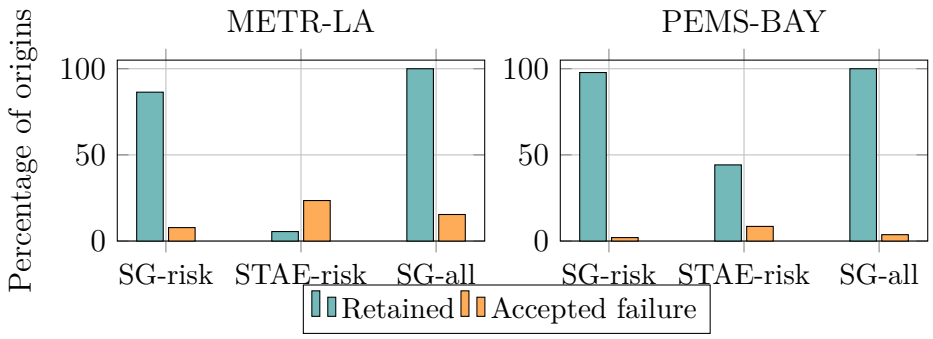


Figure 5: Operational selection outcomes at 30% severity. SG-risk denotes ShiftGuard with the registered controller; STAE-risk applies the same controller to STAEformer; SG-all releases every forecast. High retention is useful only when accepted-origin failure also remains low.

Removing structured augmentation increases MAE by 0.306 mph on METR-LA and 0.703 mph on PEMS-BAY; independent masking alone also loses 0.098 and 0.220 mph. Conversely, removing reliability-conditioned propagation improves MAE by 0.143 and 0.067 mph, and removing the router has a small favorable effect. Table 8 therefore supports structured augmentation as the effective architectural contribution but falsifies the stronger claim that every proposed mechanism is necessary.

ShiftGuard uses 89k/100k parameters on METR-LA/PEMS-BAY versus 1.26M/1.37M for STAEformer. Its batch-one latency is 2.30/2.43 ms, and its batch-eight throughput is 3567/3335 samples s^{-1}, compared with 503/269 for STAEformer on the same RTX 5060 Laptop GPU. Table 9 reports the principal local efficiency controls.

Table 8: Ablation results at 30% severity, averaged over non-clean families.

Dataset	Variant	MAE	AURC	Retained
METR-LA	ShiftGuard	3.399	2.400	0.864
METR-LA	Adaptive graph only	3.319	2.362	0.884
METR-LA	Independent masking only	3.496	2.508	0.820
METR-LA	No outage augmentation	3.705	2.657	0.740
METR-LA	No reliability conditioning	3.256	2.339	0.906
METR-LA	No regime router	3.372	2.392	0.872
METR-LA	Static graph only	3.469	2.448	0.841
PEMS-BAY	ShiftGuard	1.948	1.130	0.978
PEMS-BAY	Adaptive graph only	1.884	1.108	0.983
PEMS-BAY	Independent masking only	2.168	1.204	0.935
PEMS-BAY	No outage augmentation	2.651	1.735	0.917
PEMS-BAY	No reliability conditioning	1.881	1.098	0.979
PEMS-BAY	No regime router	1.927	1.115	0.977
PEMS-BAY	Static graph only	2.056	1.238	0.952

Table 9: Local GPU efficiency; latency uses batch one and throughput/memory use batch eight. FLOPs are profiler-observed operations per sample.

Dataset	Model	Parameters	FLOPs/sample	Latency B1 (ms)	Throughput B8	Memory B8 (MiB)
METR-LA	DCRNN	63372	845712576.00	7.97	1052.04	38.27
METR-LA	Graph WaveNet	22744	189736407.00	1.86	7136.17	47.52
METR-LA	STID	41996	4040706.00	0.47	16822.62	34.97
METR-LA	STAEformer	1258932	6080479272.00	2.62	502.59	180.93
METR-LA	ShiftGuard	89012	38541287.00	2.30	3567.45	36.95
PEMS-BAY	DCRNN	63372	1806417600.00	7.47	1045.04	42.00
PEMS-BAY	Graph WaveNet	25104	387135775.00	1.50	6071.88	57.62
PEMS-BAY	STID	49548	6344066.00	0.49	16769.73	36.60
PEMS-BAY	STAEformer	1372212	10386051000.00	3.68	268.54	319.69
PEMS-BAY	ShiftGuard	100340	76656703.00	2.43	3335.28	42.18

Figure 6 complements the error tables with resource cost. ShiftGuard is roughly fourteen times smaller than STAEformer and uses about one fifth to one eighth of its batch-eight allocated GPU memory. The advantage is especially large on PEMS-BAY, where the spatial attention axis contains 325 sensors. STID and Graph WaveNet remain faster than ShiftGuard in the local benchmark, so ShiftGuard is not an absolute efficiency SOTA. Its claim is instead a favorable joint operating point: substantially lower outage error than those compact baselines while using far fewer resources.

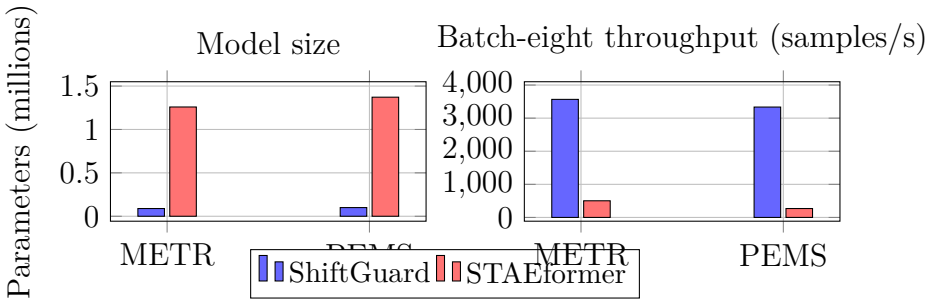


Figure 6: Efficiency comparison with the strongest local transformer. ShiftGuard uses 13.7–14.1 times fewer parameters and provides 7.1–12.4 times higher batch-eight throughput on the disclosed GPU.

7. Discussion

7.1. Answers to the research questions

RQ1: point robustness. The results support the value of structured-outage training on both datasets. Averaged over 20 non-clean family–severity scenarios, ShiftGuard reduces MAE relative to the strongest local transformer by 0.347 mph on METR-LA and 0.546 mph on PEMS-BAY. The crossed seed–scenario bootstrap intervals exclude zero, and the separation grows with severity in Figure 3. The effect is not confined to independent masking: Figure 4 shows an advantage in every registered outage family. Region failure remains hardest for ShiftGuard, because a connected missing set removes both local history and the most useful neighbors. The smaller region advantage on METR-LA (0.116 mph) identifies a specific residual weakness rather than invalidating the broader effect.

RQ2: subgroup interval reliability. At 30% severity, ShiftGuard's 90% intervals achieve about 0.898 marginal coverage on both datasets. More importantly, worst node–horizon coverage exceeds STAEformer by 0.015 on METR-LA and 0.017 on PEMS-BAY, while worst-regime coverage improves by 0.073 and 0.186. The PEMS-BAY difference is operationally meaningful: STAEformer's 0.701 worst-regime coverage means that a marginally acceptable interval can fail badly for a particular traffic state. Group-prequential calibration does not make every subgroup exactly 90%; finite calibration samples, temporal dependence, and abrupt transitions prevent that interpretation. It does, however, reduce the disparity relative to the strongest comparator.

RQ3: selective risk. The origin-level controller provides the clearest deployment advantage. On METR-LA, ShiftGuard retains 86.4% of forecasts with a 7.8% empirical failure rate, compared with 5.5% retention and 23.5% failure for STAEformer under the same selection rule. On PEMS-BAY it retains 97.8% with 2.0% failure, versus 44.2% and 8.5%. Releasing every ShiftGuard forecast raises failure to 15.4% and 3.7%, respectively. The PEMS-BAY all-forecast rate is already below the 10% target, yet risk control still removes a small harmful tail. On METR-LA, abstention materially changes whether the target is met. AURC improvements of 1.46 and 1.26 show that the interval-derived score ranks high-error origins better across the full retention continuum, not only at one threshold.

RQ4: mechanisms and cost. The ablations narrow the explanation. Removing structured augmentation increases 30%-severity MAE by 0.306 mph on METR-LA and 0.703 mph on PEMS-BAY; replacing it with independent-only masking also worsens performance. Conversely, removing reliability-conditioned graph propagation improves MAE by 0.143 and 0.067 mph, and disabling the router has a small favorable effect. The robustness gain therefore cannot honestly be attributed to every architectural module. It is primarily associated with the training distribution, calibrated scale output, and the decision layer. This negative result is scientifically useful: a lean successor may omit the router and simplify message weighting. ShiftGuard remains computationally practical, using 89k/100k parameters and 37/42 MiB batch-eight peak allocated memory. Its throughput is 7.1 and 12.4 times that of the local STAEformer on METR-LA and PEMS-BAY, respectively.

7.2. Why clean accuracy and deployment reliability disagree

The clean results illustrate why average benchmark rankings should not be treated as deployment rankings. STAEformer and ShiftGuard are indistinguishable on METR-LA at the tested resolution, while STAEformer is significantly better on clean PEMS-BAY. Under outage, the relationship reverses. STAEformer's adaptive embeddings and attention are highly effective when all node histories are present, but the model was not optimized to interpret an explicit failure mask or to reconstruct information from structured partial histories. ShiftGuard sacrifices a small amount of clean PEMS-BAY accuracy in exchange for a training distribution that covers missingness. This is a conventional robustness trade-off rather than evidence that either model is universally superior.

For an ITS operator, the preferred model depends on the cost function. If sensors are centrally managed, outages are negligible, and every prediction must be released, STAEformer may be preferred on PEMS-BAY. If communication loss, maintenance, or corridor failure is plausible and the controller can fall back to a safe policy, ShiftGuard offers a more useful risk–coverage profile. This distinction motivates reporting clean accuracy, stressed accuracy, calibration, selection, and resource cost as separate axes rather than collapsing them into one leaderboard.

7.3. Statistical interpretation

The hierarchical bootstrap resamples both seeds and outage scenarios, matching the two principal replicated axes. Its confidence intervals support a stable average robustness effect. The seed-level Wilcoxon tests remain non-significant after Holm correction because five seeds permit only a coarse attainable two-sided rank-test resolution. We therefore do not use the bootstrap interval to claim that the adjusted rank test is significant, nor do we reinterpret non-significance as equivalence. Both are reported because they answer different questions: the interval estimates the effect under hierarchical resampling, whereas the rank test provides a conservative seed-level check.

The risk certificate has an equally important qualification. The exact binomial upper bound assumes that accepted-origin failures in the certification sample behave as i.i.d. Bernoulli trials. Traffic origins overlap and are temporally dependent, so that assumption is an approximation. Disjoint construction and certification samples prevent direct outcome reuse, and the prequential order prevents future leakage, but neither operation proves independence. Accordingly, the paper calls the observed quantities empirical prequential coverage and failure. A stronger time-series certificate would require mixing assumptions, blockwise methods, martingale bounds, or online guarantees explicitly designed for dependent sequences.

7.4. Threats to validity and generalizability

Several limitations bound the claims. First, METR-LA and PEMS-BAY are freeway speed datasets from 2012 and 2017. They do not represent signalized urban arterials, multimodal demand, weather extremes, special events, or contemporary connected-vehicle telemetry. Second, the correlation graph is learned from training data rather than a verified directed road topology. Correlation can connect sensors that co-vary because of common demand but

lack a direct physical link; it also makes the region and hub stress families properties of the proxy graph. Third, synthetic outages are reproducible and controllable but cannot capture every field failure mechanism. Real outages may co-occur with incidents, produce delayed or biased readings rather than complete absence, and have repair dynamics not represented here.

Fourth, locally reproduced graph baselines are protocol-matched compact implementations, not bitwise reproductions of every official repository. Literature-reported values are therefore kept in a separate contextual table and never enter paired significance claims. Fifth, hyperparameters were selected under a finite validation budget and may favor neither model optimally. Sixth, the operational failure threshold of 5 mph and risk limit of 10% are plausible research choices, not universal policy standards. A traffic management center must select them from the consequences of false alarms, unsafe actuation, and fallback availability. Finally, all experiments use one GPU platform; latency rankings can change with compilation, precision, batch size, and edge hardware.

Despite these constraints, the design offers useful internal validity. Both datasets share the same preprocessing and split logic; masks are deterministic and paired across models; five seeds are used for stochastic models; test targets never tune the predictor or controller; all tables are generated from machine-readable artifacts; and negative ablation findings are retained. These controls make the central conclusion reproducible within the declared scope.

8. Future work

The first priority is prospective validation. A deployment study should ingest live detector health logs, communication alarms, maintenance records, weather, and incident annotations, then evaluate forecasts before outcomes are known. Such data would reveal whether the four synthetic outage families span real failure geometry and whether risk control triggers early enough for an operator to switch to a safe fallback. Evaluation should expand to signalized urban networks, arterial travel times, traffic flow and occupancy, and geographically distinct cities. Cross-city transfer would test whether structured augmentation learns a general robustness principle or benchmark-specific correlations.

The second priority is stronger dependent-data inference. Block conformal, weighted conformal, martingale, and change-point-aware methods could

replace the i.i.d. binomial approximation. A promising design would jointly adapt the rolling window, subgroup shrinkage, and risk threshold according to detected regime duration while preserving anytime validity. Hierarchical calibration could pool information among nearby sensors without allowing large groups to hide minority undercoverage. Conditional coverage remains impossible without additional assumptions in full generality, so future work should state the precise dependence or smoothness structure under which stronger guarantees are obtained.

Third, the negative ablations suggest a simpler model family. Removing the router and reliability-weighted propagation, then reallocating the parameter budget to a stronger mask-aware temporal encoder, may improve both clean and stressed accuracy. Learned corruption generators could approximate empirical failure logs, while adversarial graph outages could locate worst-case vulnerable subnetworks. Any such extension should retain the current fixed-mask paired evaluation to separate a genuinely stronger model from a more favorable stress sample.

Fourth, decision quality should be connected to transportation outcomes. Forecast MAE and a 5 mph tensor threshold are proxies. Simulation could measure travel time, queue spillback, ramp-metering stability, incident response delay, or regret of the downstream control action. Abstention could select among fallbacks rather than return a binary rejection: persistence, historical profiles, a physics-based model, human review, or a neighboring data source may have different costs. Multi-objective risk control could then balance safety, coverage, latency, and energy use.

Finally, uncertainty should be decomposed and communicated. Deep ensembles, Bayesian approximations, and conformalized quantile models could distinguish data uncertainty from model uncertainty, while attribution maps could explain which unavailable sensors caused abstention. Edge deployment studies should include quantization, mixed precision, pruning, and end-to-end communication latency. These directions would move the present benchmark contribution toward a certifiable, interpretable, and maintainable ITS service.

9. Conclusion

This study reframes short-term network traffic forecasting as a joint point, interval, and selective-decision problem under compound traffic-regime and sensor-availability shift. ShiftGuard combines an explicit observation

mask, a compact graph–temporal predictor, structured-outage augmentation, heterogeneity-aware scales, rolling node–horizon conformalization, and an origin-level act/abstain rule. The accompanying protocol evaluates independent, burst, contiguous-region, and hub failures at five severities on METR-LA and PEMS-BAY, with identical masks, five seeds, hierarchical paired inference, and separate clean and stressed claims.

The empirical conclusion is specific. ShiftGuard is not a universal clean-data SOTA: it is tied with STAEformer on METR-LA and is 0.027 mph worse on clean PEMS-BAY. Across 640 stress evaluations, however, it reduces average ShiftGuard–STAEformer MAE by 0.347 mph and 0.546 mph, improves worst-regime coverage, lowers AURC, and retains far more forecasts under the registered risk controller. It is also substantially smaller and faster than the transformer comparator. Ablation shows that structured-outage augmentation is the main architectural/training contributor, whereas reliability-weighted propagation and multi-expert routing are not necessary for the observed gain.

The defensible novelty is therefore the integrated operational formulation and evidence: a leakage-controlled compound-shift benchmark, structured-outage training, subgroup-aware prequential intervals, and tensor-level selective risk assessment. The work does not claim unconditional coverage under temporal dependence or field-wide generalization from two legacy benchmarks. Within its declared scope, it demonstrates that asking *when not to trust a traffic forecast* changes both model ranking and evaluation practice. Future progress should validate that conclusion on prospective outage logs, newer urban networks, dependent-data certificates, and downstream traffic-control outcomes.

Data and code availability

The source code, experiment configurations, fixed seeds, stress-mask generators, training and evaluation scripts, and table-generation utilities are openly available at `https://github.com/shishirbit/ShiftGuard-ITS`. Large generated checkpoints and prediction tensors are excluded from version control. The METR-LA and PEMS-BAY benchmark files are available through the official DCRNN data-preparation repository at `https://github.com/liyaguang/DCRNN#data-preparation` and its linked dataset archive at `https://drive.google.com/drive/folders/10FOTa6HXPqX8Pf5WRoRwcFnW9BrNZEIX`. PEMS-BAY originates from the

California Department of Transportation Performance Measurement System (PeMS), https://pems.dot.ca.gov/. The raw benchmark files are not redistributed in this repository.

References

[1] Y. Li, R. Yu, C. Shahabi, Y. Liu, Diffusion convolutional recurrent neural network: Data-driven traffic forecasting, in: International Conference on Learning Representations, 2018.

[2] B. Yu, H. Yin, Z. Zhu, Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting, in: Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence, 2018, pp. 3634–3640.

[3] Z. Wu, S. Pan, G. Long, J. Jiang, C. Zhang, Graph wavenet for deep spatial-temporal graph modeling, in: Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence, 2019, pp. 1907–1913.

[4] L. Bai, L. Yao, C. Li, X. Wang, C. Wang, Adaptive graph convolutional recurrent network for traffic forecasting, in: Advances in Neural Information Processing Systems, Vol. 33, 2020.

[5] C. Shang, J. Chen, J. Bi, Discrete graph structure learning for forecasting multiple time series, in: International Conference on Learning Representations, 2021.

[6] Z. Shao, Z. Zhang, W. Wei, F. Wang, Y. Xu, X. Cao, C. S. Jensen, Decoupled dynamic spatial-temporal graph neural network for traffic forecasting, Proceedings of the VLDB Endowment 15 (11) (2022) 2733–2746. doi:10.14778/3551793.3551827.

[7] C. Song, Y. Lin, S. Guo, H. Wan, Spatial-temporal synchronous graph convolutional networks: A new framework for spatial-temporal network data forecasting, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 34, 2020, pp. 914–921. doi:10.1609/aaai.v34i01.5438.

[8] J. Jiang, C. Han, W. X. Zhao, J. Wang, Pdformer: Propagation delay-aware dynamic long-range transformer for traffic flow prediction, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 37, 2023, pp. 4365–4373. doi:10.1609/aaai.v37i4.25556.

[9] R. Jiang, Z. Wang, J. Yong, P. Jeph, Q. Chen, Y. Kobayashi, X. Song, S. Fukushima, T. Suzumura, Spatio-temporal meta-graph learning for traffic forecasting, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 37, 2023, pp. 8078–8086.

[10] C. Zheng, X. Fan, C. Wang, J. Qi, GMAN: A graph multi-attention network for traffic prediction, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 34, 2020, pp. 1234–1241. doi:10.1609/aaai.v34i01.5477.

[11] S. Lan, Y. Ma, W. Huang, W. Wang, H. Yang, P. Li, DSTAGNN: Dynamic spatial-temporal aware graph neural network for traffic flow forecasting, in: Proceedings of the 39th International Conference on Machine Learning, Vol. 162, 2022, pp. 11906–11917.

[12] H. Liu, Z. Dong, R. Jiang, J. Deng, J. Deng, Q. Chen, X. Song, Staeformer: Spatio-temporal adaptive embedding makes vanilla transformer sota for traffic forecasting, in: Proceedings of the 32nd ACM International Conference on Information and Knowledge Management, 2023, pp. 4125–4129.

[13] Z. Shao, Z. Zhang, F. Wang, W. Wei, Y. Xu, Spatial-temporal identity: A simple yet effective baseline for multivariate time series forecasting, in: Proceedings of the 31st ACM International Conference on Information and Knowledge Management, 2022, pp. 4454–4458.

[14] B. N. Oreshkin, A. Amini, L. Coyle, M. Coates, FC-GAGA: Fully connected gated graph architecture for spatio-temporal traffic forecasting, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 35, 2021, pp. 9233–9241. doi:10.1609/aaai.v35i10.17114.

[15] A. Baggag, S. Abbar, A. Sharma, T. Zanouda, A. Al-Homaid, A. Mohan, J. Srivastava, Learning spatiotemporal latent factors of traffic via regularized tensor factorization: Imputing missing values and forecasting,

IEEE Transactions on Knowledge and Data Engineering 33 (6) (2021) 2573–2587. doi:10.1109/TKDE.2019.2954868.

[16] L. Deng, X. Liu, H. Zheng, X. Feng, Z. Chen, Graph-tensor neural networks for network traffic data imputation, IEEE/ACM Transactions on Networking 31 (6) (2023) 3010–3024. doi:10.1109/TNET.2023.3268982.

[17] J. Sun, Y. Fan, C.-C. M. Yeh, W. Zhang, G. Chowdhary, Revealing the power of masked autoencoders in traffic forecasting, arXiv preprint arXiv:2309.15169 (2023).

[18] H. Gao, R. Jiang, Z. Dong, J. Deng, Y. Ma, X. Song, Spatial-temporal-decoupled masked pre-training for spatiotemporal forecasting, in: Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, 2024, pp. 3998–4006. doi:10.24963/ijcai.2024/442.

[19] D. Wu, L. Gao, X. Xiong, M. Chinazzi, A. Vespignani, Y.-A. Ma, R. Yu, Quantifying uncertainty in deep spatiotemporal forecasting, arXiv preprint arXiv:2105.11982 (2021).

[20] T. Mallick, J. Macfarlane, P. Balaprakash, Uncertainty quantification for traffic forecasting using deep-ensemble-based spatiotemporal graph neural networks, IEEE Transactions on Intelligent Transportation Systems 25 (8) (2024) 9141–9152. doi:10.1109/TITS.2024.3381099.

[21] K. Stankeviciute, A. M. Alaa, M. van der Schaar, Conformal time-series forecasting, in: Advances in Neural Information Processing Systems, Vol. 34, 2021, pp. 6216–6228.

[22] I. Gibbs, E. J. Candès, Conformal inference for online prediction with arbitrary distribution shifts, Journal of Machine Learning Research 25 (162) (2024) 1–36.

[23] A. N. Angelopoulos, R. Barber, S. Bates, Online conformal prediction with decaying step sizes, in: Proceedings of the 41st International Conference on Machine Learning, Vol. 235, 2024, pp. 1616–1630.

[24] M. Patil, Q. Ahmed, S. Midlam-Mohler, Urban traffic forecasting with integrated travel time and data availability in a conformal graph neural network framework, in: 2024 IEEE 27th International Conference on

Intelligent Transportation Systems, 2024, pp. 2482–2487. doi:10.1109/ITSC58415.2024.10920016.

[25] Y. Wang, Y. Ma, Adaptive conformal prediction for spatio-temporal hypergraph neural networks, Applied Soft Computing (2026). doi:10.1016/j.asoc.2026.116256.

[26] C. Wang, Z. Wang, W. Wang, J. Gan, Y. Ju, Cold-start traffic state forecasting at unseen sensor locations via support-conditioned metagraph learning, Sensors 26 (15) (2026) 4995. doi:10.3390/s26154995.

[27] J. Wu, S. Wei, Z. Zhang, F. Wei, C. Wang, Beyond correlations: A causal invariance framework for robust urban traffic prediction, International Journal of Approximate Reasoning (2026) 109802doi:10.1016/j.ijar.2026.109802.

[28] Z. Wu, S. Pan, G. Long, J. Jiang, X. Chang, C. Zhang, Connecting the dots: Multivariate time series forecasting with graph neural networks, in: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2020, pp. 753–763. doi:10.1145/3394486.3403118.

[29] C. Cheong, G. Davis, S. Choi, Weaver: Kronecker product approximations of spatiotemporal attention for traffic network forecasting, Transportation Research Part C: Emerging Technologies (2026).

[30] K. Martin, S. Heilig, A. Fischer, M. F. C. Haddad, A. M. Sykulski, M. Eliasof, A critical audit of spatiotemporal forecasting benchmark datasets and baselines, arXiv preprint arXiv:2608.20980 (2026). doi:10.48550/arXiv.2608.20980.

[31] Z. Shao, Z. Zhang, F. Wang, Y. Xu, Pre-training enhanced spatial-temporal graph neural network for multivariate time series forecasting, in: Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2022, pp. 1567–1577. doi:10.1145/3534678.3539396.

[32] H. Lee, S. Ko, TESTAM: A time-enhanced spatio-temporal attention model with mixture of experts, in: International Conference on Learning Representations, 2024.

[33] Z. Zhang, W. Zhang, Y. Huang, K. Chen, Test-time compensated representation learning for extreme traffic forecasting, arXiv preprint arXiv:2309.09074 (2023).

[34] W. Chen, Y. Liang, Expand and compress: Exploring tuning principles for continual spatio-temporal graph forecasting, in: International Conference on Learning Representations, 2025.

[35] Y. Bao, Y. Huo, H. Ren, C. Zou, CAP: A general algorithm for online selective conformal prediction with FCR control, Journal of Machine Learning Research 26 (287) (2025) 1–74.

[36] Y. Wang, X. Sun, L. Kong, W. Gao, Y. Li, Z. Fan, Z. Zhuang, Do time series foundation model benchmarks hide regime-dependent failures? evidence from traffic speed forecasting, arXiv preprint arXiv:2606.18367 (2026). doi:10.48550/arXiv.2606.18367.